IEOR 221 Assignment 9

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1.

2. For the antithentic method, we used the opposite-sign normal samples as a pair. For the control variate method, we implemented two different control functions, which are the ternimal stock price and the price of corresponding geometric Asian call option.

Method	NPV	MC standard error
Plain MC	5.1914	0.0085
Control variate (terminal stock price)	5.1924	0.0054
Control variate (geometric Asian call)	5.1983	0.0003
Antithetic MC	5.1966	0.0069
Antithetic + control variate (terminal stock price)	5.1961	0.0046
Antithetic + control variate (geometric Asian call)	5.1987	0.0003

Each method uses 2^{20} simulated paths, the fixed seed 0, and is computed on CUDA. For antithetic methods, these paths form 2^{19} pairs. The price of the vanilla call option with the same strike and maturity is 8.8249.